



BioPlex 2200 ToRC IgG and IgM Assays

A rapid and comprehensive testing solution for ToRC IgG and IgM testing

BIO-RAD



Like No Other

The BioPlex 2200 system offers IgG and IgM assays to simultaneously detect antibodies to *T. gondii*, rubella and cytomegalovirus (CMV).

Prenatal testing for *Toxoplasma gondii*, rubella and cytomegalovirus is used to determine exposure to infectious agents that can cause congenital diseases. The BioPlex 2200 System offers a novel multiplex panel to simultaneously identify and differentiate antibodies for these infectious diseases. The BioPlex 2200 System also offers type specific HSV testing to provide a complete ToRCH panel.

The BioPlex 2200 ToRC IgG assay reports results for the following

- Anti-*T. gondii* IgG – Quantitative
- Anti-Rubella IgG – Quantitative
- Anti-CMV IgG – Qualitative

The BioPlex 2200 ToRC IgM assay reports results for the following

- Anti-*T. gondii* IgM – Qualitative
- Anti-Rubella IgM – Qualitative
- Anti-CMV IgM – Qualitative



Correlation with CDC Evaluation Serum Panels

The Centers for Disease Control (CDC) evaluation serum panels were used to evaluate the performance of the *T. gondii* IgM, *T. gondii* IgG, Rubella IgG and CMV IgG assays. The BioPlex 2200 ToRC IgM and IgG assays show excellent agreement with the CDC panel.

BioPlex 2200 Assay Results for CDC *T. gondii* IgG Reference Sera (N=100)

CDC Samples	Reference Result	BioPlex Pos (+)	BioPlex Neg (-)
<i>T. gondii</i> IgG	Pos (+)	70	0
	Neg (-)	0	30

Note: Tested at Bio-Rad Laboratories

BioPlex 2200 Assay Results for CDC *T. gondii* IgM Reference Sera (N=97)

CDC Samples	Reference Result	BioPlex Pos (+)	BioPlex Neg (-)
<i>T. gondii</i> IgM	Pos (+)	32	0
	Neg (-)	0	65

Note: Tested at Bio-Rad Laboratories

BioPlex 2200 Assay Results for CDC Rubella IgG Reference Sera (N=100)

CDC Samples	Reference Result	Site 1		Site 2		Site 3	
		BioPlex Pos (+)	BioPlex Neg (-)	BioPlex Pos (+)	BioPlex Neg (-)	BioPlex Pos (+)	BioPlex Neg (-)
Rubella IgG	Pos (+)	82	0	82	0	82	0
	Neg (-)	0	18	0	18	0	18

Note: Tested at 3 clinical evaluation sites

BioPlex 2200 Assay Results for CDC CMV IgG Reference Sera (N=100)

CDC Samples	Reference Result	BioPlex Pos (+)	BioPlex Neg (-)
CMV IgG	Pos (+)	66	0
	Neg (-)	0	34

Note: Tested at Bio-Rad Laboratories



Precision

Reproducibility near the cut-off is essential for providing consistent patient results. Reproducibility testing was performed according to the principles described in the Clinical Laboratory Standards Institute (CLSI) guidances EP5-A2 and EP5-A3.

ToRC IgG

Low Positive Sample	Sample (N)	Mean (AI)	Within-Run		Between-Run		Total	
			SD	%CV	SD	%CV	SD	%CV
<i>T. gondii</i>	60	21 IU/mL	0.95	4.5%	0.44	2.1%	1.92	9.1%
Rubella	60	13 IU/mL	0.73	5.8%	0.18	1.4%	0.90	7.2%
CMV	60	1.9 AI	0.09	4.7%	0.05	2.7%	0.21	11.3%

ToRC IgM

Low Positive Sample	Sample (N)	Mean (AI)	Within-Run		Between-Run		Total	
			SD	%CV	SD	%CV	SD	%CV
<i>T. gondii</i>	120	1.3 AI	0.05	4.0%	0.03	2.5%	0.16	12.7%
Rubella	120	1.2 AI	0.06	4.8%	0.01	1.7%	0.16	13.3%
CMV	120	1.2 AI	0.07	6.2%	0.03	2.5%	0.12	10.6%

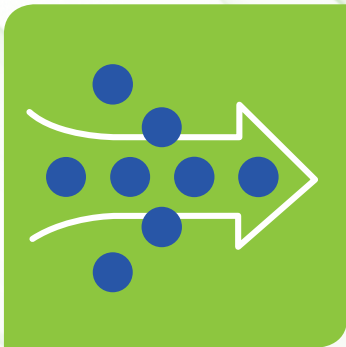
Guidelines for Interpretation of Results

<i>T. gondii</i> IgG	Rubella IgG	CMV IgG	ToRC IgM	Result
≤ 9 IU/mL	≤ 7 IU/mL	≤ 0.8 AI	≤ 0.8 AI	Negative
10 – 11 IU/mL	8 – 9 IU/mL	0.9, 1.0 AI	0.9, 1.0 AI	Equivocal
≥ 12 IU/mL	≥ 10 IU/mL	≥ 1.1 AI	≥ 1.1 AI	Positive



Efficiency

- Multiplex your workflow – consolidating six traditional assays from the ToRC panel into two convenient multiplexed reagent packs
- Multiplex your inventory – dramatically reducing reagent and calibrator storage requirements
- Multiplex your samples – significantly reducing your sample volume requirements
- Multiplex your tubes - accomodating serum or plasma (heparinized or EDTA)



Throughput

- Market-leading throughput with up to 300 multiplexed test results per hour to meet heavy workload demands
- Random access processing ensures results are available when you need them
- Time to first result of 45 minutes maximizes speed and flexibility in result reporting



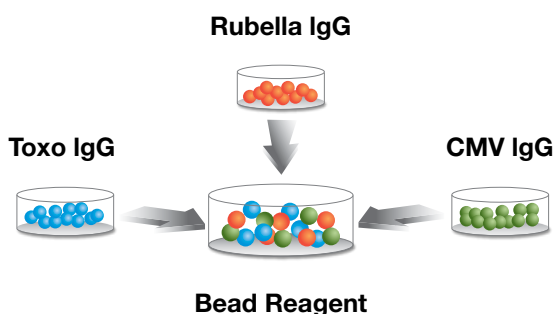
Confidence

- Full automation from primary tube to final result, minimizing the chance of human error
- Internal quality control beads included with each sample to verify serum addition and standardize detector performance
- One system with an expanding menu to meet your autoimmune and serology testing needs

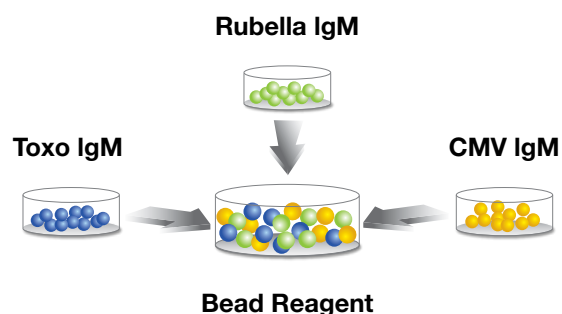
Assay Principle

The BioPlex 2200 ToRC IgG and ToRC IgM assays are multiplexed bead based immunoassays designed to detect and differentiate antibodies to *T. gondii*, rubella, and CMV. Each bead reagent is a mixture of disease specific dyed microparticle beads. The beads are individually manufactured and combined with internal QC beads into a bead reagent for multiplex analysis.

ToRC IgG Multiplex Testing



ToRC IgM Multiplex Testing



ToRC IgG
Reagent Pack

ToRC IgM
Reagent Pack



ToRC IgG
Calibrator Set

ToRC IgM
Calibrator Set



ToRC IgG
Control Set

ToRC IgM
Control Set

ToRC IgG and IgM Reagent Pack

The ToRC IgG and IgM reagent packs contain all the necessary reagents (sample diluent, beads and conjugate) to process 100 samples per pack for ToRC IgG and 150 samples per pack for ToRC IgM.

ToRC IgG and IgM Calibrator Set

Calibrators are ready to use and multiplexed. For *T. gondii* IgG and Rubella IgG, a six point curve is used to calculate quantitative results. For CMV IgG, a four point curve is used to calculate qualitative results. For all IgM assays, a two point curve is used to calculate qualitative results.

ToRC IgG and IgM Control Set

Quality Controls are ready to use and multiplexed. The ToRC IgG set includes a negative control as well as two different levels of multi-analyte positive controls. The ToRC IgM set includes one multi-analyte positive control and multi-analyte negative control.

Ordering Information

Catalog No.	Description	Catalog No.	Description
665-1650	BioPlex 2200 ToRC IgG Reagent 100 Test Pack 1 pack	12000670	BioPlex 2200 ToRC IgM Reagent 150 Test Pack 1 pack
663-1600	BioPlex 2200 ToRC IgG Calibrators 1 set	12000677	BioPlex 2200 ToRC IgM Calibrators 1 set
663-1630	BioPlex 2200 ToRC IgG Controls 2 sets	12000678	BioPlex 2200 ToRC IgM Controls 2 sets

BioPlex is a trademark of Bio-Rad Laboratories, Inc. in various jurisdictions.



**Bio-Rad
Laboratories**

For further information, please contact the Bio-Rad office nearest you or visit our website at www.bio-rad.com/diagnostics

**Clinical
Diagnostics Group**

Website www.bio-rad.com/diagnostics **Australia** 61-2-9914-2800 **Austria** 43-1-877-8901 **Belgium** 32-9-385-5511 **Brazil** 5521-3237-9400 **Canada** 1-514-334-4372 **China** 86-21-61698500 **Czech Republic** 420-241-430-532 **Denmark** +45-4452-1000 **Finland** 358-9-804-22-00 **France** 33-1-47-95-60-00 **Germany** +49-(0)89-318-840 **Greece** 30-210-7774396 **Hong Kong** 852-2789-3300 **Hungary** +36-1-459-6100 **India** 1800-180-1224 **Israel** 972-3-9636050 **Italy** +39-02-216091 **Japan** 81-3-6361-7070 **Korea** 82-2-3473-4460 **Mexico** +52 (55) 5488-7670 **The Netherlands** +31-318-540666 **New Zealand** 64-9-415-2280 **Norway** 47-23-38-41-30 **Poland** 48-22-3319999 **Portugal** 351-21-472-7700 **Russia** 7-495-721-14-04 **Singapore** 65-6415-3170 **South Africa** 27-11-442-85-08 **Spain** 34-91-590-5200 **Sweden** 46-8-555-127-00 **Switzerland** 41-61-717-95-55 **Taiwan** 866-2-2578-7189 **Thailand** 662-651-8311 **United Kingdom** +44-(0)20-8328-2000

Printed on recycled paper with soy-based inks

© 2017 Bio-Rad Laboratories, Inc. Printed in USA 6/17 DG17-0138 T-152